

■ LegendreP

`LegendreP`[n , x] gives the Legendre polynomial $P_n(x)$.

`LegendreP`[n , m , x] gives the associated Legendre polynomial $P_n^m(x)$.

Mathematical function (see Section ??). ■ Explicit polynomials are given for integer n and m . ■ The Legendre polynomials satisfy the differential equation $(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n + 1)y = 0$. ■ The Legendre polynomials are orthogonal with unit weight function. ■ The associated Legendre polynomials are defined by $P_n^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_n(x)$. ■ For arbitrary complex values of n, m and z , `LegendreP`[n , z] and `LegendreP`[n , m , z] give Legendre functions of the first kind. ■ See pages 360 and 363. ■ See also: `SphericalHarmonicY`.